Kris Ko

k9ko@uwaterloo.ca | linkedin.com/in/kris-ko | github.com/MyeongcheolKo

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering, Honours — *Expected May 2030*

Experience

Embedded Flight Software Engineer Intern — May 2026 – Aug 2026
Waterloo Aerial Robotics Group — *Waterloo, ON*

- Extended ZeroPilot, a **FreeRTOS C++ flight controller on STM32H7**, from fixed-wing to quadrotor, reaching stable hover across **10+ flight tests** after software-in-the-loop (SITL) validation.
- Built a dual ICM-42688 IMU driver on a shared SPI bus using a **ping-pong DMA state machine** that merges both FIFOs into one time-ordered stream, and fixed an attitude lag bug caused by a hardcoded dt fed into the Mahony filter.
- Root-caused ICP-20100 barometer **init failures on cold boot** (never under the debugger) to the datasheet's single dummy I2C wake-up write assuming a ready sensor; resolved by polling wake-up and version reads until a valid ID returned.
- Cut AHRS EKF runtime from **20 ms to 340 μs (59x)** by exploiting block sparsity with CMSIS-DSP 3x3 matrix operations and compiler optimization, meeting the 1 kHz control loop deadline; verified with a custom RTOS profiler.
- Decoupled SD-card logging from the real-time control loop with a message queue drained by a **low-priority preemptible thread**, eliminating control-task stalls from blocking FatFS writes.
- Traced corrupted IMU readings in **custom PCB bring-up** to a floating chip-select using oscilloscope and logic analyzer.
- Rewrote the u-blox GPS driver to auto-configure legacy M8Q and newer M10 receivers via ACK-verified UBX commands with NMEA fallback, plus a bounds-checked, corruption-safe parser fed by **UART DMA idle-line detection**.
- Implemented **self-leveling cascaded attitude control** with a 100 Hz angle P loop feeding a 1 kHz rate PID loop, motor mixing with desaturation that preserves attitude authority at high throttle, and a DShot ESC driver.
- Designed an **altitude Kalman filter** fusing accelerometer, barometer, and GPS with outlier gating and recovery, feeding a 3-stage cascaded altitude controller with LiDAR terrain following; validated in SITL.

Software & Electrical Lead — Sep 2022 – May 2025
MATE ROV Competition — *Halifax, NS*

- **Led 10+ students** building an underwater ROV's power distribution and Ethernet/UDP thruster control between the surface station and vehicle.

Projects

KoRTOS | *C, ARM Assembly, Cortex-M4* — Dec 2025 – Sep 2026

- Built a **preemptive, priority-based RTOS kernel** from scratch on STM32F446RE, with PendSV context switching on per-task PSP stacks, 1 ms time slicing among equal-priority tasks, and a swappable port layer isolating all assembly.
- Cut worst-case priority-inversion blocking from **20.3 ms to a flat 257 μs** across 1–20 ms interfering tasks by implementing mutexes with chained priority inheritance, alongside counting semaphores and ISR-safe message queues.
- Eliminated a **queue wake-up race** where a higher-priority task or ISR could claim a slot before a woken task resumed, causing stale reads or overwrites, by re-checking on wake and re-blocking for only the remaining timeout.
- Developed a **register-level GPIO/SPI/I2C/USART/CAN HAL** with no vendor code, integrating CAN with the kernel via ISR-to-queue RX and mutex-protected TX mailboxes.
- Wrote custom **linker script and startup code** from scratch; verified the kernel and HAL with 24 sample apps.

SPI-Controlled PWM Peripheral | *Verilog RTL, cocotb* — Jun 2026

- Implemented an **SPI mode-0 peripheral in Verilog** for 16 PWM channels, synchronizing SPI inputs into the 10 MHz clock domain with 2-FF synchronizers and committing writes only on nCS rising after a full 16-bit frame.
- Wrote a **self-checking cocotb testbench** sweeping duty cycles to verify PWM frequency and duty within ±1%.

Delirium Guardian | *IoT, ESP32, Wi-Fi* — Sep 2025 – Dec 2025

- Built an **ESP32-C6 wearable** streaming pulse-oximeter, motion, and temperature data to a web dashboard that scores delirium risk, batching 20 Hz I2C samples into 1 Hz Firestore uploads to reduce Wi-Fi overhead.

Technical Skills

Languages: Embedded C, C++, Python, ARM/RISC-V Assembly, Verilog, MATLAB
Firmware/RTOS/Controls: STM32 (ARM Cortex-M), ESP32 microcontrollers; bare-metal drivers, interrupt handling (ISRs), DMA, NVIC, timers, startup code, linker scripts; hardware bring-up and debugging from schematics and datasheets; FreeRTOS, real-time scheduling, RTOS internals, IPC; PID control, EKF/Kalman filtering, sensor fusion, CMSIS-DSP
Interfaces: I2C, SPI, CAN, UART, GPIO, PWM, ADC, Ethernet/UDP
Tools: Git, GDB, JTAG/SWD, Oscilloscope, Logic analyzer, STM32CubeIDE, CMake, GoogleTest, GitHub Actions, Linux